A Teaching Guide to

"Multiscale Physics of Atomic Nuclei from First Principles"

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What this guide covers:

This document walks you through the main ideas, methods, and discoveries of the paper in an accessible way. It is designed for a reader with some physics background (undergraduate or early graduate level) who wants to understand what this work accomplished and why it matters.

Prepared as an educational companion document.

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1 The Big Picture: What Problem Does This Paper Solve?

Atomic nuclei are remarkably complex quantum systems. They display physics at *many different energy scales simultaneously*—hence the word "multiscale" in the title.

Key Idea

A medium-mass nucleus has a total binding energy of roughly 8 MeV per nucleon (hundreds of MeV total), yet its lowest rotational excitation can be a *fraction of an MeV*—sometimes just tens of keV. Capturing both extremes from the same underlying theory has been one of nuclear physics' grand challenges.

Consider ^{34}Mg: its first rotational excitation is only 0.26% of its total binding energy. Imagine trying to compute a skyscraper's height to millimetre accuracy—and then, with the *same calculation*, resolving the thickness of a coat of paint on the roof. That is the scale separation these authors confront.

1.1 Why is this hard?

There are two fundamentally different kinds of correlations in a nucleus:

1. **Short-range (dynamic) correlations.** These involve pairs or triples of nucleons interacting at close range. They account for the lion's share of the binding energy and can be captured systematically by methods like coupled-cluster theory, with computational cost scaling polynomially with mass number.
2. **Long-range (static) correlations.** These involve *many* nucleons moving collectively—the nucleus deforming, rotating, vibrating. They contribute comparatively little binding energy but are essential for understanding the nucleus' shape and electromagnetic properties.

Analogy

Think of an orchestra. The "short-range correlations" are like the individual musicians playing their parts correctly (the fundamental skill). The "long-range correlations" are the collective swell of a crescendo—an emergent property that arises from many musicians coordinating. You need both to describe the performance, but they require very different descriptions.

Previous *ab initio* ("from first principles") calculations could do one or the other well, but not both simultaneously with full nuclear forces. This paper achieves that.

2 Background Concepts

2.1 From QCD to Nuclear Forces

The fundamental theory of the strong force is **quantum chromodynamics (QCD)**, which describes quarks and gluons. However, QCD cannot be solved analytically at the low energies relevant to nuclear structure. Instead, physicists use **chiral effective field theory (EFT)**—a systematic approximation that describes interactions between nucleons (protons and neutrons) in terms of pion exchanges and short-range contact terms, organized by a power-counting scheme:

Order	**Abbreviation**
Leading order	LO
Next-to-leading order	NLO
Next-to-next-to-leading order	NNLO
Next-to-next-to-next-to-leading order	$\mathrm{N^3LO}$

Each order introduces more terms with adjustable parameters called **low-energy constants (LECs)**. The paper uses interactions at NNLO (with explicit Δ isobars) and $\mathrm{N^3LO}$ (for the nucleon-nucleon part of the 1.8/2.0(EM) interaction).

2.2 What is *ab initio* nuclear theory?

An *ab initio* calculation starts from:

1. A Hamiltonian derived from chiral EFT (encoding the nuclear force).
2. A many-body method to solve the quantum mechanical problem for A interacting nucleons.

No phenomenological parameters (like "effective charges") are fitted to the nuclei being studied. The goal is *predictive power*: if the forces are right, the nuclear properties should emerge.

2.3 Nuclear Deformation and Rotation

Most nuclei are not perfect spheres. Open-shell nuclei (those with partially filled proton or neutron shells) tend to deform into prolate (rugby-ball) or oblate (frisbee-like) shapes. A deformed nucleus can *rotate*, and the resulting energy spectrum follows the rigid-rotor pattern:

$$E(J) = \frac{J(J+1)}{2\,\Theta}, \tag{1}$$

where J is the angular momentum quantum number and Θ is the moment of inertia. A hallmark of a good rigid rotor is the energy ratio $R_{42} \equiv E(4^+)/E(2^+) \approx 10/3 \approx 3.33$.

The strength of the electric quadrupole ($E2$) transition from the first 2^+ state to the 0^+ ground state, denoted $B(E2;\downarrow)$, directly measures the degree of collectivity and deformation.

2.4 Shape Coexistence

Some nuclei can exist in two (or more) shapes at nearly the same energy—for instance, a near-spherical configuration and a strongly deformed one. This phenomenon, called **shape coexistence**, is especially interesting near "magic numbers" (shell closures), where the competition between spherical shell effects and deformation-driving correlations is most acute.

3 The Theoretical Framework

The authors' strategy can be summarized in three layers, each addressing a different energy scale.

3.1 Layer 1: The Deformed Reference State (Hartree–Fock)

The starting point is a **Hartree–Fock (HF)** calculation. Each nucleon is treated as moving independently in an average potential created by all the others. By allowing the solution to break rotational symmetry, one obtains a deformed Slater determinant $|\Phi_0\rangle$—typically with prolate (elongated) shape.

The Hamiltonian used is:

$$H = T_{\text{kin}} - T_{\text{c.m.}} + V_{NN} + V_{NNN}, \tag{2}$$

where T_{kin} is kinetic energy, $T_{\text{c.m.}}$ subtracts centre-of-mass motion, V_{NN} is the two-nucleon force, and V_{NNN} is the three-nucleon force.

Takeaway

The deformed HF state is the "leading-order" description of the nucleus. It already contains the essential physics of deformation and sets the stage for everything that follows.

3.2 Layer 2: Short-Range Correlations (Coupled-Cluster Theory)

On top of the HF reference, the authors apply **coupled-cluster (CC) theory**, which parameterizes the many-body wave function as:

$$|\Psi\rangle = e^{T}|\Phi_0\rangle, \qquad T = T_1 + T_2 + \cdots + T_A, \tag{3}$$

where T_n creates n-particle–n-hole excitations. In practice, the series is truncated at doubles (CCSD: $T \approx T_1 + T_2$), capturing about 90% of the correlation energy. Including leading triples corrections brings this to ∼99%.

Key Idea

Coupled-cluster theory is *size-extensive* (the energy scales properly with particle number) and scales polynomially ($\sim N^6$ to N^7) with system size. This makes it practical for medium-mass nuclei. It captures the "high-momentum", short-range correlations that account for the bulk of the binding energy.

The authors use a **bivariational** formulation, where the left (bra) and right (ket) states are parameterized independently:

$$|\Psi\rangle = e^{T}|\Phi_0\rangle, \tag{4}$$

$$\langle\tilde{\Psi}| = \langle\Phi_0|(1+\Lambda)\,e^{-T}. \tag{5}$$

This is essential for computing transition matrix elements consistently.

3.3 Layer 3: Long-Range Correlations (Angular-Momentum Projection)

The deformed coupled-cluster state $|\Psi\rangle$ is a superposition of states with different angular momenta. To extract states with definite spin J, one applies the **angular-momentum projection operator**:

$$P^{J} = \frac{2J+1}{2}\int_0^{\pi} d\beta\; d^{J}_{00}(\beta)\; R(\beta), \tag{6}$$

where $R(\beta) = e^{-i\beta J_y}$ rotates the state by angle β, and d^J_{00} is a Wigner d-function. The projected energy is:

$$E(J) = \frac{\langle \tilde{\Psi} | P^J H | \Psi \rangle}{\langle \tilde{\Psi} | P^J | \Psi \rangle}. \tag{7}$$

Common Misconception

A naïve implementation of this projection (the "disentangled" approach of Qiu et al.) can lose accuracy for large rotation angles and produce kernels that lack the correct $\beta \to \pi - \beta$ symmetry. The authors develop a **new method** (Eq. (6) in the paper) that integrates over an auxiliary parameter λ at fixed angle β, dramatically improving numerical stability and symmetry properties.

3.4 Combining the Layers

The full calculation pipeline is:

Spherical HF (for normal-ordering 3N forces)
↓
Deformed HF (axially symmetric reference $|\Phi_0\rangle$)
↓
Coupled-Cluster (CCSD: dynamic correlations → binding energy)
↓
Angular-Momentum Projection (static correlations → rotational spectra, $B(E2)$)

4 Uncertainty Quantification

A distinctive feature of this work is its serious treatment of uncertainties, which come from three sources:

1. **EFT truncation error.** The chiral expansion is truncated at a finite order. Higher-order terms are estimated to contribute ∼10% relative uncertainty.
2. **Many-body method error.** Truncating coupled-cluster at CCSD and the projection at 2p–2h introduces errors estimated at 5–15% depending on the observable.
3. **Model-space truncation.** Finite basis sizes (8–13 harmonic-oscillator shells) introduce a controllable error, gauged by varying the oscillator frequency $\hbar\omega$.

For systematic uncertainty propagation, the authors use an ensemble of 100 calibrated interactions from chiral EFT with Δ degrees of freedom at NNLO (ΔNNLO). These were previously calibrated against light nuclei and ^{28}O data. By performing the full calculation for each interaction sample and applying **Bayesian importance resampling** (using the 2^+ and 4^+ energies of ^{24}Ne as calibration data), they generate posterior predictive distributions for all observables, with 68% and 90% credible intervals.

Takeaway

This Bayesian approach allows the authors to make predictions with quantified error bars—a major step beyond single-interaction "point predictions." The resulting uncertainties are on par with experimental uncertainties for neutron-rich neon isotopes.

5 Key Results

5.1 Rotational Bands and $B(E2)$ in $^{20–34}$Ne

The paper's headline result is shown in Figure 3 of the original: for even-mass neon isotopes from ^{20}Ne to ^{34}Ne, the computed excitation energies $E(2^+)$ and $E(4^+)$ and the transition strengths $B(E2;\downarrow)$ agree with all available experimental data within the theoretical error bars.

Key observations:

- For $N \geq 20$ (i.e., ^{30}Ne and beyond), the spectra follow a rigid-rotor pattern with large moments of inertia, indicating strong deformation.
- The computed R_{42} for ^{34}Ne is 3.37 ± 0.13 (posterior), very close to the rigid-rotor value of 10/3.
- ^{32}Ne and ^{34}Ne are predicted to be strongly deformed and collective. Their $B(E2)$ values and structure are genuine predictions for future experiments.

5.2 Shape Coexistence in ^{30}Ne

A particularly exciting prediction is **shape coexistence in ^{30}Ne**. This nucleus has the "magic" neutron number $N = 20$, which in the traditional shell model favours a spherical shape. However, the authors find *two* energy minima in constrained Hartree–Fock calculations:

Configuration	Deformation β_2	Relative Energy
Near-spherical	0.10	+1.8 MeV above deformed
Prolate deformed	0.44	Ground state

The deformed ground state signals the breakdown of the $N = 20$ magic number in this neutron-rich region—the so-called "island of inversion." This has profound implications for our understanding of the nuclear drip line and the structure of ^{28}O and its neighbours.

5.3 What Drives Nuclear Deformation?

Using **eigenvector continuation** (a reduced-order model technique), the authors build fast emulators for projected Hartree–Fock calculations, enabling a **global sensitivity analysis** with one million samples of the 17 low-energy constants.

The main findings from this variance-based analysis:

- More than 90% of the variance in R_{42} is explained by first-order (main) effects.
- About 40% of the deformation is driven by two parameters: the subleading pion-nucleon coupling c_3 and the subleading singlet S-wave contact $C_{^1S_0}$.
- Increasing short-range repulsion (via S-wave contacts) tends to *increase* deformation, likely by suppressing pairing.
- Increasing medium-range two-pion exchange strength also increases deformation, by adding attraction in higher partial waves.
- For neutron-rich nuclei (^{32}Ne, ^{34}Mg), the isospin-breaking neutron-neutron S-wave contact becomes more important.

Key Idea

This is the first time anyone has quantitatively traced nuclear deformation back to specific terms in the chiral EFT Hamiltonian. It bridges the gap between the high-resolution picture (chiral forces) and the low-resolution shell-model intuition ("pairing vs. quadrupole interactions").

6 Technical Innovations

6.1 New Angular-Momentum Projection Method

The original disentangled approach solves:

$$e^{V(\beta)}e^{T} = e^{W_0(\beta)+W_1(\beta)+W_2(\beta)+\cdots}$$

by differentiating with respect to β and integrating from $\beta = 0$. This suffers from stiffness when $\langle\Phi_0|R(\beta)|\Phi_0\rangle \approx 0$.

The new method instead introduces $e^{\lambda V}e^{T} = e^{W_0(\lambda)+W_1(\lambda)+W_2(\lambda)+\cdots}$ and differentiates with respect to λ at *fixed* β. Integration from $\lambda = 0$ to 1 is numerically stable, preserves $\beta \to \pi - \beta$ symmetry, and gives more accurate results for all angles.

6.2 Reduced-Order Model for Hartree–Fock

To emulate projected HF calculations across millions of LEC values, the authors use eigenvector continuation. They expand the target HF state in a "snapshot basis" of 68 HF solutions computed at carefully chosen LEC values:

$$|\phi_\star\rangle \approx \sum_{i=1}^{68} c_i|\phi_i\rangle.$$

A critical subtlety: the Hamiltonian in the subspace eigenvalue problem must be the *Hartree–Fock Hamiltonian* (not the full Hamiltonian), ensuring the solutions remain Slater determinants rather than superpositions thereof.

6.3 $B(E2)$ from Bivariational Coupled-Cluster

Computing electromagnetic transitions in coupled-cluster theory requires care because the left and right states differ. The authors derive expressions involving similarity-transformed quadrupole operators and rotation operators, benchmarking against the symmetry-adapted no-core shell model to validate their approach.

7 Broader Significance

7.1 For Nuclear Physics

- This work provides the first *ab initio* description that simultaneously captures binding energies (short-range physics) and rotational spectra with $B(E2)$ transitions (long-range physics) using realistic nuclear forces including three-nucleon interactions.
- The prediction of shape coexistence in ^{30}Ne and strong deformation in ^{32,34}Ne provides concrete targets for experiments at rare-isotope facilities (FRIB, RIKEN, etc.).
- The sensitivity analysis provides the first quantitative link between chiral EFT parameters and emergent nuclear deformation.

7.2 For Many-Body Theory

- The new angular-momentum projection method for coupled-cluster states is directly applicable to quantum chemistry, where symmetry-broken reference states are also common.
- The reduced-order model for non-linear eigenvalue problems (HF) using eigenvector continuation is a methodological advance applicable to any mean-field calculation with parametric Hamiltonians.
- The framework of "size-extensive method + symmetry projection" provides a conceptually clean multiscale decomposition applicable to other finite fermion systems (molecules, atomic clusters).

8 Glossary of Key Terms

Ab initio
: From first principles; no phenomenological adjustments to the nuclei being studied.

Chiral EFT
: Effective field theory of nuclear forces organized by a power-counting scheme based on pion masses and nucleon momenta.

Coupled-cluster theory
: A many-body method using an exponential ansatz $e^T|\Phi_0\rangle$; size-extensive and polynomially scaling.

CCSD
: Coupled-cluster singles and doubles—the workhorse truncation of CC theory.

Angular-momentum projection
: Mathematical operation that extracts states of definite spin J from a deformed (symmetry-broken) state.

$B(E2)$
: Reduced electric quadrupole transition probability; measures collectivity and deformation.

R_{42}
: Ratio $E(4^+)/E(2^+)$; equals 10/3 for a perfect rigid rotor.

Shape coexistence
: Existence of two or more nuclear shapes at similar energies in the same nucleus.

Island of inversion
: Region of the nuclear chart (around $N = 20$, neutron-rich) where the expected shell closure breaks down and deformation dominates.

Eigenvector continuation
: A reduced-basis method that uses solutions at known parameter values to emulate solutions at new parameter values.

LECs
: Low-energy constants—the free parameters of chiral EFT, determined by fitting to data.

Normal ordering
: Rewriting the Hamiltonian relative to a reference state, separating zero-, one-, and two-body contributions.

Drip line
: The boundary of nuclear existence; the most neutron-rich (or proton-rich) isotope that is bound.

9 Summary: The Multiscale Picture

The Conceptual Framework in One Box

Energy scales in a nucleus:

∼940 MeV	⟵	Nucleon mass
∼140 MeV	⟵	Pion mass
∼8 MeV/nucleon	⟵	Binding energy
∼0.1–1 MeV	⟵	Rotational excitations
∼0.01–0.1 MeV	⟵	Heavy-nucleus rotations

How this paper captures them:

Physics	Method	Energy fraction
Nuclear force	Chiral EFT	Sets the scale
Mean field	Deformed HF	∼85% of binding
Dynamic correlations	Coupled-cluster (CCSD)	∼14% of binding
Triples correction	CCSDT-1	∼1% of binding
Static correlations	Angular-momentum projection	$< 1\%$ of binding, but drives all collective physics

Takeaway

The main message of this paper is both simple and profound: the symmetry-breaking reference state contains the essential deformation physics. Short-range correlations (coupled-cluster) provide the binding energy. Long-range correlations (projection) contribute little energy but are responsible for the rich collective structure—rotational bands, electromagnetic transitions, and shape coexistence—that defines how nuclei behave. All of this emerges from nuclear forces rooted in QCD, with quantified uncertainties, opening the door to understanding how the most basic features of the strong force sculpt the shapes of atomic nuclei.